Supplementary Technical Appendix: Avatar Animation

1 Overview

This appendix describes the algorithm used to generate the avatar's facial animation during the bargaining task. The goal of the design is to produce naturalistic, time-continuous facial behavior.

The avatar's facial expressions are generated as the combination of (i) spontaneous expressions and (ii) mimicry responses to externally inferred affective signals. All stochastic components are pre-specified and do not depend on participant actions.

2 Smile Generation Model

At each time t, the avatar's smile intensity $S(t)$ is defined as:

$$S(t) = \max\{S_{\text{base}}, S_{\text{spont}}(t), S_{\text{mimic}}(t)\},$$

where $S_{\text{base}} = 0.15$ is a constant baseline expression (neutral but 'alive').

Spontaneous expressions. Mild spontaneous smiles were animated on the avatars during bargaining rounds. This serves two purposes. First, they provide a baseline level of facial activity that is independent of the mimicry mechanism, ensuring that any observed effects are not driven by the mere presence of smiling, but by the contingent nature of the avatar's responses.

Second, they ensure that participants are exposed to a minimal level of affiliative signaling, which increases the likelihood that facial expressions emerge during the interaction. This is important because mimicry responses require (by definition) observable affective input; without such baseline activation, mimicry dynamics would be weak or absent for some participants in the experimental condition.

Importantly, spontaneous smiles are pre-specified and identical across experimental conditions, and therefore do not introduce systematic differences between treatment groups.

Spontaneous smiles are generated by a trigger process producing smooth rise–hold–fade dynamics. Each trigger has:

- Rise duration: 0.75 ± 0.25 seconds,
- Hold duration: 2 ± 0.75 seconds,
- Fade duration: 3 ± 1 seconds,

with peak intensity drawn uniformly from $[0.2, 0.4]$. These parameters generate temporally smooth, moderate-intensity expressions. The total number of spontaneous smiles was held fixed across bargaining rounds, with 7 spontaneous smiles spaced throughout the bargaining window of 60 seconds. The shape of each spontaneous smile had random variability following the previously defined margins.

Mimicry component. Mimicry responses are driven by an externally inferred affective signal (DDAMFN smile probability; Zhang et al., 2023). The signal is smoothed using an exponential moving average (EMA) with two time scales:

$$\tau_{\text{fast}} = 0.65 \text{ s}, \quad \tau_{\text{slow}} = 3.5 \text{ s}.$$

A fixed latency of 0.5 seconds is introduced between signal detection and avatar response. In the control condition, the mimicry component is bypassed altogether.

3 Stochastic Modulation

Two stochastic processes introduce variability in the avatar's responsiveness:

- **Ornstein–Uhlenbeck process:** governs time-varying responsiveness with parameters $\mu = 1$, $\theta = 0.01$, $\sigma = 0.05$, and values constrained to remain within ± 0.3 around the mean.
- **Bounded random walk:** modulates mimicry intensity within the interval $[0.6, 1]$, updated as a small random walk at each time step.

These processes ensure that avatar behavior remains dynamic and non-deterministic while preserving smooth temporal evolution.

4 Temporal Smoothing

The mimicry signal is updated using an exponential moving average:

$$\mathrm{EMA}_t = \mathrm{EMA}_{t-1} + \alpha(\tilde{S}(t) - \mathrm{EMA}_{t-1}),$$

where $\tilde{S}(t)$ denotes the incoming affective signal and the update rate $\alpha = 1 - e^{-\Delta t/\tau}$ varies across fast and slow time scales. This prevents abrupt changes and produces continuous facial motion.

5 Additional Animation Components

Facial animation is augmented with pre-recorded motion data, including eye movements, blinking patterns, and head motion. Blinks follow a fixed temporal structure consisting of:

- Closing phase: 0.10 seconds,
- Hold phase: 0.04 seconds,

- Opening phase: 0.16 seconds.

These components are replayed continuously and are independent of participant behavior. Two motion data schedules were pre-recorded (one per bargaining round against the avatar) and were held fixed across participants.

6 Exogeneity

All components of the animation process are pre-specified and do not respond to either the participant's or the avatar's bargaining behavior or outcomes. However, we refrain from claiming strict exogeneity in advance, as the treatment will naturally exhibit some degree heterogeneity across treated participants by the very definition of the phenomenon being studied.